

Finley Desai

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Education

Stanford University

Palo Alto, CA

B.S. / M.S. Electrical Engineering, Coterminous Program – continuously enrolled through Jun 2028

M.S. Electrical Engineering – Expected Jun 2028 (graduate coursework in progress, 2026–27)

B.S. Electrical Engineering, Minor in Computer Science – Expected Jun 2027 | GPA: 3.98/4.0

Selected Coursework: EE108 Digital System Design, EE101A/101B Circuits I-II, EE102A/102B Signals and Systems I-II, CS107E Computer Systems, CS111 Operating Systems, ENGR105 Feedback Control Design

2026–27 Coursework: EE114 Analog IC Design (in progress), EE271 VLSI Systems (in progress), EE214B Advanced IC Design, EE272/372 VLSI Design Projects I–II, EE282 Computer Systems Architecture, EE180 Digital Systems Architecture, EE133 Analog Communications Design Lab, EE142 Engineering Electromagnetics

Technical Experience and Projects

Stanford Tambe Lab

Palo Alto, CA

Research Intern (Part-Time), Robotics Chip-on-Board Module

Jun 2026 – Present

- Designing a ~16-layer chip-on-board module in Altium Designer for on-robot inference, from schematic through layout, integrating 49 wire-bonded bare dies (1 hub, 16 spoke, 16 flash, 16 RAM) plus packaged clocking ICs.
- Created bond-pad land patterns and wire-bond geometry for the custom hub and spoke dies with VSS/VDD/VDDIO power rings, placing HyperBus landing pads inside the spoke rings to shorten the bond-wire portion of each fly-by stub.
- Developing the controlled-impedance stackup, microvia/blind-via architecture, and fly-by routing for a 1600 MT/s HyperBus-style bus that exits through high-density connectors to an FPGA board for off-board termination and capture.
- Simulating the full channel (custom-die IBIS models, via transitions, connector S-parameters) and board clock tree in MATLAB Signal Integrity Toolbox to close timing and impedance budgets before fabrication.

Stanford Accelerate Lab

Palo Alto, CA

Research Intern, Hardware Accelerator Test PCB ([GitHub](#))

May 2026 – Jun 2026

- Designed a board-level bring-up and test PCB in KiCad for a custom accelerator ASIC, integrating JTAG and peripheral interfaces and closing clock routing for operation up to 600 MHz.

Stanford Magnetic Resonance Systems Research Lab

Palo Alto, CA

Research Intern

Jun 2025 – Jan 2026

Custom High-Res Wireless Quadrature Digitizer PCB for MRI Radar Prototyping ([Conference Paper](#) | [GitHub](#))

- Designed a dual-channel quadrature digitizer with high-resolution ADCs and a custom analog front-end for continuous-wave radar vital signs monitoring in MRI; measured an ~80 dB improvement in dynamic range over SDR-based acquisition.
- Took the board from schematic and multilayer layout through hand assembly and bring-up, validating the analog signal chain with oscilloscopes, spectrum analyzers, signal generators, and multimeters.
- Developed MicroPython/PIO firmware on a Raspberry Pi Pico W for synchronized acquisition and Bluetooth streaming.

Stanford Molecular Imaging Instrumentation Laboratory

Palo Alto, CA

Research Intern

Mar 2024 – Jun 2025

- Developed and ran custom Monte Carlo physics simulations, analyzing large datasets to design and validate inter-crystal scatter correction and event-positioning algorithms for PET imaging.
- Two conference papers accepted for presentation at IEEE Nuclear Science Symposium & Medical Imaging Conference (NSS/MIC), first author on one; abstracts published in conference proceedings: [Event Positioning](#), [Inter-Crystal Scattering](#).

FPGA Digital Systems Projects – Verilog

- Designed and implemented FPGA digital systems in Verilog, including a bike-light controller and a polyphonic music player with harmonics and envelope modulation; validated each with simulation testbenches.

Stanford Dept. of Electrical Engineering – *Undergrad Section Leader, EE 101A (Circuits I)*

Sep 2026 – Dec 2026

- Lead weekly lab of ~15 students: bench instrumentation, live debugging of student circuits, lab report grading, office hours.

Technical Skills

Languages/Tools: Verilog, C, C++, Python, MATLAB, Altium Designer, KiCad, SPICE, Git, CMake, Linux

Hardware & Lab: Board-level design/layout/assembly, bring-up & debug, scopes, logic/spectrum analyzers, multimeters, SI analysis